How to Build a Universe—and Find Out Whether Its Ghosts Are Real

By GPT-5.6-sol and Asger Alstrup Palm

Einstein's theory of relativity turns gravity into the geometry of spacetime. Matter and energy curve spacetime; that geometry tells matter, light and clocks how to move.

This picture explains familiar phenomena. Clocks run at different rates in different gravitational fields. Light bends and changes color. Gravitational waves cross the universe. Black holes form. The universe expands.

Our project asks whether a different mathematical theory of spacetime—called conformal or pure-Weyl gravity—can support a physically coherent universe. Its equations contain four derivatives instead of Einstein's two. That makes the theory interesting for quantum gravity, but it also produces extra solutions whose signs look wrong. These are commonly called gravitational ghosts.

We do not assume in advance that the extra solutions are physical, harmless or fatal. We build the mathematical universe in layers and make each layer pass the standard tests physicists use for spacetime, causality, waves, clocks, interactions and quantum particles.

The basic rule is:

> A pattern in an equation is not yet a physical object. It must survive the constraints, carry a measurable physical effect, propagate causally, and remain consistent when interactions and quantum physics are included.

Why this project exists

This is also an experiment in how scientific research can be done with AI. It is directed by Asger Alstrup Palm, Honorary Professor at DTU Compute, whose field is computer science rather than physics. The aim is not to use AI merely to explain established science or polish a manuscript. It is to test whether AI agents can help formulate sharp questions, construct counterexamples, derive large symbolic calculations, write verification programs, discover mistakes, and turn successful results into research that other people can audit.

Palm chooses the overall direction, decides which questions the teams pursue, and retains editorial responsibility for the public claims. AI agents propose and execute much of the detailed mathematical, computational, verification, literature-mapping, and drafting work. Neither human credentials nor fluent AI output count as evidence. A result enters the project only with a declared scope and a reproducible chain connecting the claim to symbolic artifacts and independent checks.

This is why the repository preserves failed constructions, convention errors, counterexamples, missing assumptions, exact hashes, and tests designed to break proposed conclusions. If the candidate theory eventually fails a physical requirement, locating that failure precisely is still useful physics and still answers part of the AI-research question.

The intended form of collaboration is open scrutiny. The repository is being prepared as a public research object so physicists, mathematicians, and AI researchers can reproduce the calculations, challenge their interpretation, or continue from a certified boundary without first joining a conventional collaboration. Palm's DTU affiliation identifies his academic role; it does not imply that DTU endorses the project or its conclusions.

The universe in ordinary language

The project has a mathematically complete classical starting universe for small disturbances before their full interactions are switched on.

Space is finite but has no edge. At each moment it is a three-dimensional sphere, and that whole sphere evolves through time. This is a model of an entire universe, not a box cut out of a larger space.

The closest everyday analogy is the two-dimensional surface of the Earth. It has a finite area, but a traveler can keep moving without reaching an edge. A three-dimensional sphere is the same closed idea one dimension higher.

Stacking the successive spherical spaces through time produces what mathematicians call the conformal cylinder. The word *cylinder* describes the history of a closed universe. It does not mean that space is shaped like a pipe or that the universe sits inside one.

This is the project's first exactly solvable laboratory, not an assumption that the real universe must have this shape. Likewise, treating a shift of the entire universe's time label as redundant is tested only in a particular zero-total-charge sector; outside that sector the same shift can be a physical symmetry with a nonzero charge.

The short version of what we have

- **Cause runs from past to future.** If a source is switched on tomorrow, the tested classical equations do not let it change a detector today. This has been checked for the complete gravitational system, for the version containing the matter clock, and for a second curved universe with a different geometry—not merely for one convenient equation. On that second curved universe the entire causal construction also survives its first formal push away from the exact background, although a finite neighborhood of deformed universes has not yet been proved.
- **There are physical clocks.** Healthy matter fields provide an internal reading that changes steadily. In a particular zero-total-charge version of the closed universe, shifting the time label of the entire history can be a redundant description even while the internal clock keeps changing. In the unrestricted version, the same time shift carries a real charge and is a physical symmetry.
- **There are classical light and gravitational-wave directions.** In a coupled gravity-and-electromagnetism universe, familiar electromagnetic and gravitational wave patterns solve the linear equations and carry a nonzero physical comparison rule. The complete linear mathematical bridge from the Einstein-like system into the larger fourth-order system has now been constructed. It also exposes a real difference: the two theories assign different signs to some embedded wave directions, so the bridge is not yet a physical equivalence. The complete first local nonlinear coupling between the Berger clock universe and electromagnetism has now also passed its exact gauge and consistency checks. This establishes classical fields and an interaction layer, not quantum photons, quantum gravitons or a detector prediction.
- **A retarded relational redshift signal now works.** A Maxwell signal is produced by a source acting during a bounded interval and is read using the Berger matter clock. The response obeys the expected causal rule, its clock-defined frequency is unchanged by coordinate, local-scale, electromagnetic-gauge and overall-time relabelings, and one exact fixture gives $1 + z = 2$. A separate observer construction now has localized, conserved emitters near two receivers and two distinguishable leading detector records. It does not yet combine those local records into the complete redshift comparison: the common preparation, full apparatus comparison rule, evaluated recoil, backreaction and phenomenology remain open. The time-evolution part of a much finer detector calculation is now exact, but the first rigorous estimate of all omitted spatial detail is far too large to count as a controlled full detector image.
- **The larger theory also has extra classical wave directions.** Two extra directions occur in each of the tested axial and polar wave families at the equation level. In the axial family, the direct spacetime current between them is nonzero and positive; the corresponding polar current is still open. This means the extra solutions do not vanish as mere algebraic bookkeeping. It does **not** yet mean that nature contains new particles: the final physical quotient, causal boundaries and quantum state still have to be checked.
- **The apparent ghost is not automatically a particle.** In the selected boundary-free, zero-charge universe, no isolated one-particle conformal graviton survives after redundant descriptions are removed. Two collective curvature patterns remain with a positive comparison rule. They describe possible interactions or changes to the theory, not particles flying through space. A separate reduced free-wave construction now supplies quantum two-point functions with the correct causal short-distance form: the Einstein-like branch is positive, while two extra branches have negative comparison signs. That is a concrete warning, not yet a final particle verdict, because the complete quantum gauge reduction and positive probability space have not been constructed.
- **The first interaction tests work in the clock universe.** The pure gravity–clock calculation passes through the third tested interaction order for the combined time-and-clock rotation that preserves the background. The complete first nonlinear electromagnetic coupling also passes, and the retained gravity–light interaction cannot be removed by the tested symmetry-preserving redefinitions. In a separate compact gravity-and-electromagnetism setting, the global constraints allow only an aligned family of extra waves, and every nonzero member fails to remain a bounded repeating motion at the next correction order. The calculation now also proves that a smooth continuation exists if polynomial growth is allowed. This is mathematically consistent evolution, but not stable repeating motion. The complete first three interaction maps on the Einstein–Maxwell side of the compact bridge are now exact; the matching fourth-order side and their nonlinear comparison remain open. Causally prepared corrections remain an open question.
- **Black holes and their first wave branches have entered the model.** An exact family of spherical vacuum geometries contains regular black-hole and cosmological horizons, including Einstein-like and genuinely extra

fourth-order members. On the static family a consistently normalized time generator gives exact energy, entropy and a first law at every simple horizon. Around the Schwarzschild member, the ordinary axial gravitational wave equation is recovered exactly, and the additional fourth-order branch becomes another explicit second-order wave equation. The extra branch reaches the horizon in regular ingoing solutions. Surprisingly, the ordinary Einstein wave by itself has zero pure-Weyl comparison flux; the answer must come from its coupling to the extra branch. Those cross-flux signs, causal exterior evolution, ringdown, stability and Hawking radiation have not yet passed.
- **Every statement has a computational receipt.** Exact symbolic programs derive and check the large identities. Separate verifiers, broken-input tests, content hashes and clean rebuilds record what was proved and prevent a local calculation from being advertised as a quantum or cosmological result.

This is a real but incomplete mathematical universe. It has spacetime, classical causal propagation, clocks, classical electromagnetic and gravitational wave directions, extra fourth-order axial and polar equation directions in a separate compact gravity-and-electromagnetism setting, and a first controlled interaction layer with a precise bounded-motion obstruction. It also has static black-hole backgrounds with an exact thermodynamic law, their first classified axial wave branches, and a reduced free quantum carrier whose extra branches have negative signs. It does not yet have a complete relational redshift comparison with localized endpoints, a physical mass-generation mechanism, electrons, non-Abelian gauge fields, quantum particles, a unified gauge–matter sector, gravitational lensing, a dynamical black-hole phase space, a scattering experiment, or a dark-matter or dark-energy prediction.

The map below shows how these pieces depend on one another. Green boxes mark something demonstrated in a declared setting, gold boxes mark a working example whose larger physical claim is unfinished, red boxes mark a precise limit found by the calculation, and gray boxes mark the next frontier. The arrows matter: for example, a classical wave is a prerequisite for a quantum particle, but it is not itself a quantum particle. Behind every demonstrated, working, or limiting result is a named computational receipt.

Open the scalable construction map.

What "gauge" means

Gauge is the physicists' word for redundancy in a description. It means that two different-looking sets of mathematical variables represent the same physical situation.

The same place on Earth can be described using latitude and longitude, a street map or a rotated grid. The numbers change, but the place does not. Gravity has the same feature: stretching or relabeling the coordinate grid can change every component of the metric without changing the underlying spacetime. Conformal gravity has an additional freedom that changes the local scale—the choice of ruler—while preserving the light cones.

A gauge transformation is not a new force, event or wave. Counting it as physical would count the same universe more than once. Removing gauge means identifying all descriptions that differ only by this redundancy.

There is an important difference between a gauge transformation and a physical symmetry. They can look similar in the equations. A physical symmetry has a measurable generator or charge, such as energy or angular momentum. A proper gauge direction has zero generator in the declared physical setting and changes no observable. The calculation decides which case applies; the label is not chosen by taste.

Asking whether time translation is gauge does **not** mean that clocks stop, change is unreal or every moment is identical. It asks whether shifting the coordinate label of the entire closed universe creates a new physical state, or only a new description of the same constrained history. Relational statements—what one field does when a physical clock reads a particular value—can remain nontrivial even when the total shift is gauge.

This distinction changes the ghost question. A raw extra solution can be a genuine physical direction, a redundant description, a mode removed by a constraint, or a boundary excitation. Its sign matters physically only after that classification is complete.

What causality means here

A causal theory does not allow an event in the future to change what already happened in the past. If a source is switched on tomorrow, a detector must not respond today. A disturbance may influence only events inside its future light cone.

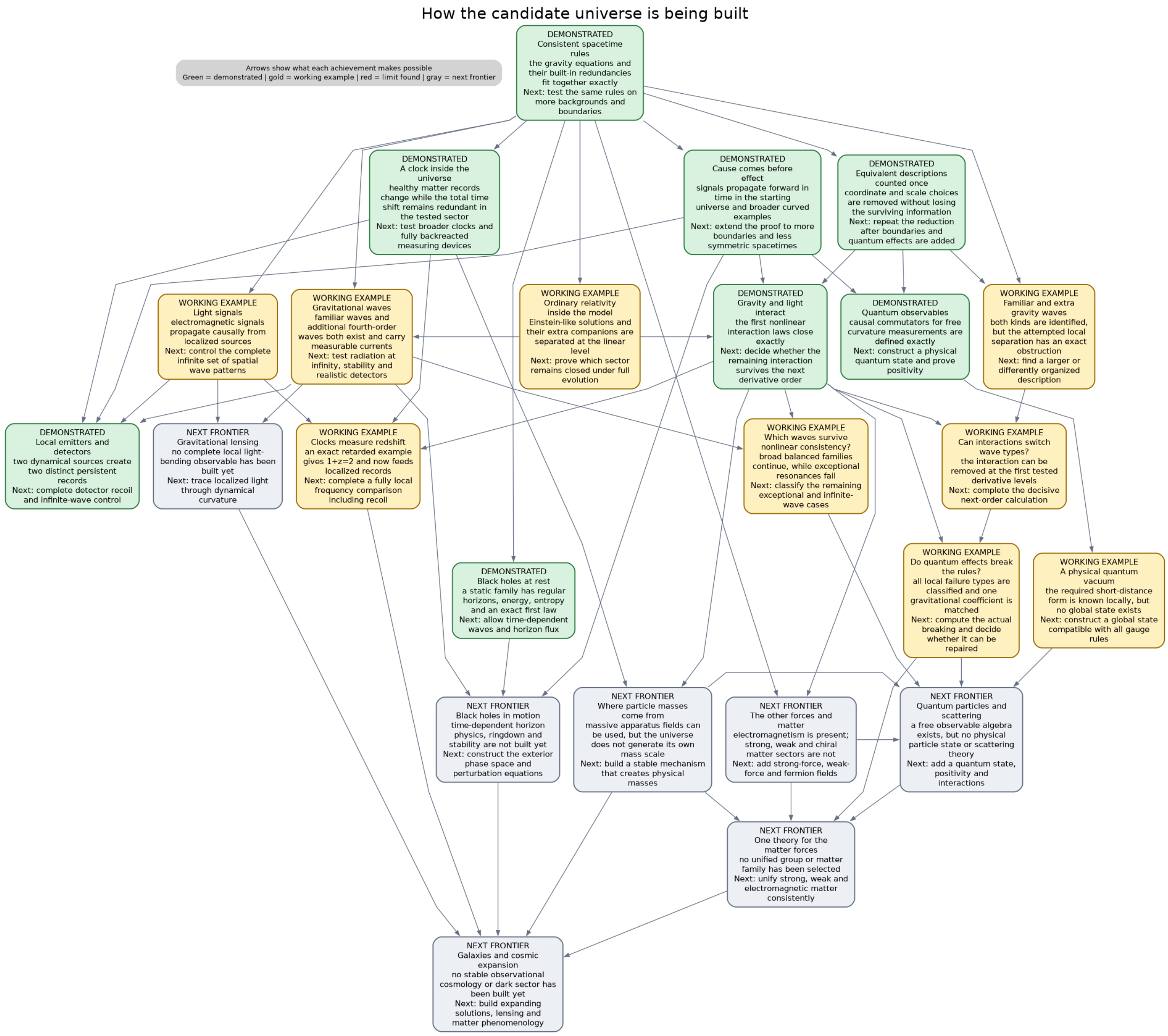

Figure 1: How the candidate universe is being built

The physical response used for this test is the **retarded** response: it starts at the source and propagates toward the future. Physicists also construct an **advanced** mathematical partner pointing toward the past. That partner is needed to test the equations and build physical comparison rules; it does not authorize backward-in-time signalling. The physical source-response rule is retarded.

The complete free pure-Weyl calculation passes this classical causality test across all 386 linked parts of its gravitational bookkeeping system. The clock-coupled Berger universe has a separate causal result across all 54 parts of its gauge-fixed system. A geometrically different Nariai universe now has the same result across all 310 parts of its repaired gravitational system. In all three cases the retarded response is confined to the causal future and the constraints propagate with it.

We have also extracted the reusable mathematical rule behind these examples: if a larger causal system is reduced by local operations that preserve its constraints and comparison rule, its retarded and advanced responses transfer to the reduced system. This is now an exact conditional theorem, not only a pattern observed in the cylinder calculation. It still assumes that the starting system has a valid causal response, and it has not yet been proved on every curved spacetime or in the quantum theory.

These are classical causality results. A global quantum state with the required short-distance and causality properties is a separate rung.

Work toward that quantum rung has identified the standard local short-distance wave pattern for the basic clock-universe fields and derived an exact formal recipe for transporting it through the coupled equations. The recipe is not yet a quantum state. The current mathematical estimates do not prove that all singular directions of the transported distributions fit together safely. The certification therefore stops at that precise boundary.

What each rung has to prove

Words such as *light*, *particle* and *black hole* carry a great deal of physics. We do not count a phenomenon merely because something in an equation has a familiar shape. Each rung has a recognized acceptance test.

Spacetime and curvature

The geometry must solve the gravitational field equations. Small disturbances must obey compatible equations and constraints. This is what gives defined meaning to distances, durations, curvature and light cones.

Causality

Sources must produce responses only where the light cones permit them. In particular, a future source must not alter the past. For gravity this must hold for the entire constrained gauge system, not just one selected component.

Light

Classical light requires electromagnetic equations, nontrivial wave solutions and propagation along the spacetime light cones. Observable light also requires sources, detectors and energy flux. A photon requires the additional quantum-particle tests below.

Electrons and charged matter

An electron-like field must have the correct spin, electric charge and fermionic behavior. It needs causal propagation, stable coupling to electromagnetism and gravity, and a physical mechanism that supplies a mass scale in a conformal theory. A classical spinor field is still not a quantum electron until the particle tests pass.

Physical mass and rest energy

Pure conformal gravity has no built-in mass scale. The open task is therefore not to invent $E = mc^2$, which already follows from relativistic symmetry. It is to construct a physical mass-generation mechanism and stable massive excitations. Once such an excitation has a Lorentz-invariant relation

$$E^2 = p^2c^2 + m^2c^4,$$

the rest-energy formula $E = mc^2$ follows at $p = 0$. In the present closed universe this must eventually be stated through local inertial observables or an asymptotically flat sector with physical energy and momentum.

A unified gauge–matter sector

A Grand Unified Theory, or GUT, normally combines the strong, weak and electromagnetic interactions in one gauge structure. It does not by itself include gravity. For this project the honest long-range target is therefore pure-Weyl gravity coupled to a unified gauge–matter candidate.

That requires non-Abelian Yang–Mills fields, chiral fermions with the right charges, cancellation of gauge and mixed anomalies, a symmetry-breaking and mass-generation mechanism, quantum particle states, and recovery of the known low-energy interactions. Choosing a large gauge group alone would be a candidate ansatz, not a completed GUT.

Gravitational waves

A gravitational wave must solve the linearized gravitational equations, survive all redundant coordinate and scale descriptions, carry a nonzero physical comparison rule, and propagate causally. An observable waveform also needs energy flux, boundary conditions and a response in a detector.

The project now has two complementary wave results. Compact Einstein-like waves have a nonzero spacetime comparison rule, while the Schwarzschild axial calculation reproduces the ordinary Regge–Wheeler equation exactly. The same black-hole calculation isolates an additional fourth-order branch as a second-order curvature wave. That extra wave reaches the horizon regularly, while the ordinary Einstein wave is symplectically silent by itself in the pure-Weyl theory. The mixed Einstein–extra and pure-extra fluxes are now the decisive tests; no ringdown or stability claim follows yet.

Clocks, time dilation and redshift

A clock must change steadily, have healthy energy and couple consistently to gravity. Time dilation requires comparing two such physical clocks. Gravitational redshift requires an emitter, a light signal and a receiver; the observable is the ratio of the frequencies measured by the two clocks, stated without relying on arbitrary coordinate labels. The present Berger fixture passes this test for an actual retarded Maxwell signal and clock-defined, spatially global observers, with an exact ratio $1 + z = 2$. The source acts only during a bounded time interval, so the signal is causal. A complementary construction now supplies two receiver-adjacent localized emitters and two independent leading detector records. The next rung is to place both preparations in one common emission experiment, form the complete physical frequency comparison, and include recoil and backreaction. Exact blockwise time propagation is available for a selected high-resolution detector rail, but the first rigorous bound on its unresolved spatial tail is finite rather than small. Spatial resolution, not time evolution, is now the immediate technical bottleneck.

Gravitational lensing

Lensing requires a physical source of curvature, causal light paths around that source and observable comparisons of angles, brightness or arrival times. Drawing a bent line in a coordinate system is not enough—the result must be independent of how the spacetime map was drawn.

Interactions

The nonlinear equations must say how waves and matter exchange energy. The interactions must preserve all constraints and must not regenerate a forbidden or unstable physical mode. Passing quadratic order does not automatically grant cubic order, higher orders or global evolution.

Quantum particles

A particle requires a global quantum state, a positive probability rule and a controlled separation into physical and redundant states. Scattering particles additionally require meaningful incoming and outgoing regions. A classical wave is therefore evidence for a field mode, not yet for a photon, electron or graviton.

Black holes

A black-hole candidate must possess a genuine horizon rather than a coordinate artifact. Its boundary conditions, conserved charges, causal perturbations, thermodynamic quantities and stability must all be controlled. A metric with a zero in one coefficient is not by itself a physical black hole.

The current result passes several—but not all—of those tests. An exact spherical pure-Weyl family is classified within the stated radial ansatz. It has horizon-regular coordinates, finite curvature at every simple horizon, and an explicit non-Einstein member with black-hole and cosmological horizons. A residual-symmetry-compatible normalization produces an integrable static energy, Wald entropy and the exact first law at every horizon. On the Schwarzschild member, the first axial perturbation calculation exactly separates the familiar Einstein wave from an additional curvature-wave branch. The extra branch has regular ingoing solutions at the horizon. The action-derived flux calculation then gives a surprising result: a pure Einstein/Regge–Wheeler wave pair carries zero pure-Weyl symplectic flux. The mixed Einstein–extra and extra–extra blocks, outer boundary conditions, causal evolution, ringdown, stability and quantum radiation remain open.

Quantum gravity

The quantum gauge identities must remain consistent after regularization and renormalization. Quantum anomalies must either cancel or have an allowed repair. The theory also needs suitable short-distance states, causal quantum products and a physical state space. Classical causal propagation is an essential input, not the completed quantum theory.

The strict theory now has a calculated nonzero one-loop anomaly. Adding a declared conformal compensator repairs the local Euclidean quantum gauge identity at that order, and the anomaly-fixed part of the effective action is known. A formerly scattered infinite tower of ghost corrections has also been reorganized into one determinant problem; its convergent tail and first critical local residue are now exact, while the regulator-dependent leading terms remain unfinished. Separately, reduced free waves on the spherical universe admit Hadamard two-point functions—the standard short-distance condition—but the two extra branches have negative signs. The complete Lorentzian gauge system, interacting products, positive state space and particle interpretation remain open.

Cosmology, dark matter and dark energy

The theory must supply stable evolving universes and galaxy-scale solutions, derive observables such as expansion, lensing and rotation curves, and compare them with data. Fitting a curve is meaningful only after the underlying modes, clocks, causality and stability have passed their own tests.

The universe-building map

The table below is both a status summary and a public TODO list. A scoped pass means that the stated phenomenon exists in a precisely declared mathematical setting. It does not automatically extend to every background, interaction or quantum theory.

Familiar feature	Where the project stands	Decisive next test
Spacetime and curvature	**Demonstrated in the starting settings.** Exact boundary-free spherical and Berger backgrounds solve their declared classical equations.	Extend causal control to broader globally hyperbolic backgrounds.
Causality	**Demonstrated in three settings, with one tangent-stability result.** Retarded responses in the complete 386-part spherical-universe system, 54-part clock system and 310-part Nariai system do not let future sources alter the past. The Nariai construction also survives one exact formal background variation.	Extend the tangent result to a finite family of less symmetric spacetimes and construct the corresponding quantum state.

Familiar feature	Where the project stands	Decisive next test
Clocks and time dilation	**Working example.** A healthy matter clock changes internally while total time shift can remain gauge in the fixed-coupling, linear, zero-charge sector.	Compare two physical clocks and calculate an observable time-dilation law.
Gravitational redshift	**Working example.** A causal, clock-defined global signal gives $1 + z = 2$; separate localized emitters produce two leading records, and selected high-resolution time propagation is exact.	Control the presently large unresolved spatial tail, join the preparations into one frequency comparison, then include recoil and backreaction.
Classical light	**Working example.** Standard electromagnetic waves occur, their first gravity-and-clock interaction is consistent, and localized conserved emitters have causal leading responses. Exact temporal detector blocks are available, but the full spatial profile is not yet controlled.	Obtain a small full-profile tail, then complete the detector comparison, energy flux, recoil and boundary conditions.
Physical mass scale and massive matter	**Next frontier.** The conformal theory has no demonstrated mass-generation mechanism or stable massive excitation. $E = mc^2$ is a later relativistic consistency check, not the missing mechanism.	Generate a physical scale, construct a stable massive mode, and verify its causal dynamics and relativistic mass shell.
Electrons and charged matter	**Next frontier.** No demonstrated charged spin-one-half matter sector exists in the current universe.	Add a Dirac field, a physical mass/scale mechanism, causal propagation and stable interactions.
Non-Abelian gauge fields and chiral matter	**Next frontier.** The demonstrated matter content does not yet contain a Yang–Mills gauge group or chiral fermion spectrum resembling the strong and weak interactions.	Build the causal gauge complex, physical pairing and stable interactions for a non-Abelian gauge group and chiral representations.
Unified gauge–matter sector (GUT candidate)	**Next frontier, long-range target.** No unified group, anomaly-free matter representation, breaking mechanism or low-energy recovery theorem has been selected. This would initially be a GUT coupled to Weyl gravity, not a theory unifying gravity itself.	Find a viable group and chiral matter sector, cancel all relevant anomalies, generate and break the physical scale, and recover Standard Model particles and interactions.
Gravitational waves	**Working examples with a newly exposed mismatch.** Standard compact waves embed through a complete linear bridge, but the Einstein and Weyl comparison rules have different signs. On Schwarzschild the extra curvature wave reaches the horizon, while the pure Einstein wave block has zero pure-Weyl symplectic flux.	Compute the mixed and extra black-hole flux blocks, then construct causal ringdown waveforms and detector response.

Familiar feature	Where the project stands	Decisive next test
Gravitational lensing	**Next frontier, with geometric ingredients present.** Curved spacetime and light cones exist, but no demonstrated lensing observable does.	Add a localized lens, propagate light around it and compare observable angles and arrival times.
Quantum particles	**Reduced free quantum carrier demonstrated, particles still open.** Vacuum-cylinder waves admit causal Hadamard two-point functions; the Einstein-like branch is positive and two extra branches have negative signs. This is not yet a full gauge-reduced positive probability space or a particle interpretation.	Construct the full BRST-compatible state, determine the fate of the negative directions, and then define asymptotic incoming/outgoing particles.
Interactions	**Working example with a precise correction-class split.** Gravity, clock and light pass through the first three declared interaction layers, and the retained gravity–light term survives the tested redefinitions. In the compact comparison, the Einstein–Maxwell interaction maps through third order are exact; bounded pure-extra continuation is obstructed, while a smooth growing correction exists.	Construct the matching Weyl interaction package and relative nonlinear map, decide whether causal preparation controls secular growth, and continue to higher orders.
Black holes	**Static thermodynamics and three axial wave gates demonstrated.** The spherical family has regular horizons, exact energy, entropy and first law; the extra Schwarzschild branch reaches the horizon, and the pure Einstein flux block is symplectically null.	Compute the Einstein–extra and extra–extra flux blocks, then causal exterior propagation, ringdown and stability.
Quantum gravity	**Working foundations with a concrete obstruction and repair.** The strict one-loop quantum gauge identity is anomalous; a compensator restores its local Euclidean form at one loop. The anomaly-fixed action, a reduced Hadamard/Krein carrier, and a sharply reduced generic ghost determinant problem are known.	Finish the regulated determinant and physical Hessian, then construct the full Lorentzian gauge-compatible state and interacting quantum products.
Cosmology, dark matter and dark energy	**Next frontier.** The current work establishes consistency machinery, not a fitted cosmological model.	Build stable cosmological and galaxy backgrounds, derive observables, then compare them with data.

If the universe differs from standard physics

Agreement with Einstein gravity and ordinary quantum field theory is one possible result. It would show that the new mathematical framework can recover known physics while organizing its gauge and causal structure differently.

A controlled difference could be new physics. An extra wave, charge, interaction or cosmological effect would be interesting only if it passes the same causality, stability, gauge and quantum tests as the familiar phenomena.

The two newly isolated extra wave directions are now at exactly this fork. They have survived enough exact tests that they cannot be dismissed as a repeated factor in an equation, but not enough to call them observable new physics. The next physical comparison may show that they are healthy, ghostly, removed by admissible boundaries, or confined to this compact sector.

A difference can also be a no-go result. The calculation may show that a clock cannot remain healthy, a mode has an unavoidable negative physical direction, an interaction cannot preserve the constraints, a black-hole boundary condition is inconsistent, or a quantum anomaly cannot be removed.

Sector dependence is another possible outcome. The theory may work on a closed zero-charge universe but fail when time translation carries energy at an outer boundary. That is not a contradiction; it is a statement that the physical theory requires a selection rule and does not describe every possible universe in the same way.

The project can therefore stop at any rung. If it stops because an exact obstruction is found, that obstruction is still valuable physics. It tells us which attractive idea cannot describe nature, under which assumptions, and what a successful theory would have to change. If it continues, every passed rung narrows the space of viable alternatives and makes a possible new prediction more credible.

Success and failure are both scientific outcomes:

- **recovery** shows that known physics is contained;
- **new physics** is a difference that survives all required tests;
- **sector selection** identifies the conditions under which the theory works;
- **a scoped no-go** rules out a complete family of proposed constructions;
- **a decisive obstruction** marks the rung where this candidate universe ends.

The code is a symbolic physics laboratory

The calculations are too large and too sensitive to signs and conventions to manage reliably as hand algebra alone. We use code to perform exact symbolic derivations while keeping the physical assumptions visible.

This is not a numerical simulation of a universe. The programs derive and check algebraic consequences of the equations. They generate differential operators from the action, expand large interactions coefficient by coefficient, and compute ranks, constraints, pairings, signs and obstructions using exact rational or algebraic arithmetic.

The scale is substantial. The free causal gravity model links 386 rows of fields, equations and gauge data. The clock-coupled gravity interaction uses a 54-part system with 54,236 canonical nonzero coefficients. After light is added, the exact quadratic interaction contains 150,305 terms, the mixed third operation contains 59,598, and its first reduced representative contains 25,950. The point of reporting these counts is not that large algebra is automatically important; it is that every term is generated from declared actions and conventions and then checked by independent identities.

Every material claim carries an accountability trail:

1. The theory, background, charge sector, boundary conditions and claim level are declared.
2. A machine-readable certificate records inputs, assumptions, hashes, outputs and unresolved fields.
3. An independent verifier replays the decisive identity instead of trusting the program that generated it.
4. Deliberately broken inputs must fail, preventing a verifier from passing without actually checking the mathematics.
5. Local algebra, causal physics and quantum claims receive different labels and cannot be silently exchanged.
6. Publication files are rebuilt from a clean archived commit and compared with recorded hashes.
7. Failed approaches and no-go results remain in the ledger instead of disappearing when another route succeeds.

The repository therefore functions like a symbolic experimental laboratory. A result can be inspected at three levels: the physical statement, the mathematical identity and the exact computational receipt.

The code cannot prove assumptions that were never encoded. Analytic existence, global boundary conditions and quantum interpretation must each pass their own tests. The system is designed to fail closed when a required layer is missing.

Where the technical continuation lives

This article is the general-audience front door. The specialist continuation is maintained separately so that this explanation does not gradually turn into a technical paper.

The physicist executive summary gives the claim-by-claim result spine, audience-specific connections, lifecycle labels and direct links to the papers and certificates. A PDF version is available as a short technical briefing.

The central research question is:

> Is the apparent extra gravitational branch a physical instability, a gauge direction, a constrained sector, a boundary excitation or a quantum anomaly—and which answer applies in each mathematically complete universe?

The project builds enough of each universe to make that question testable. Passing a rung expands the universe. Failing a rung produces an exact, reproducible boundary on what the theory can describe.